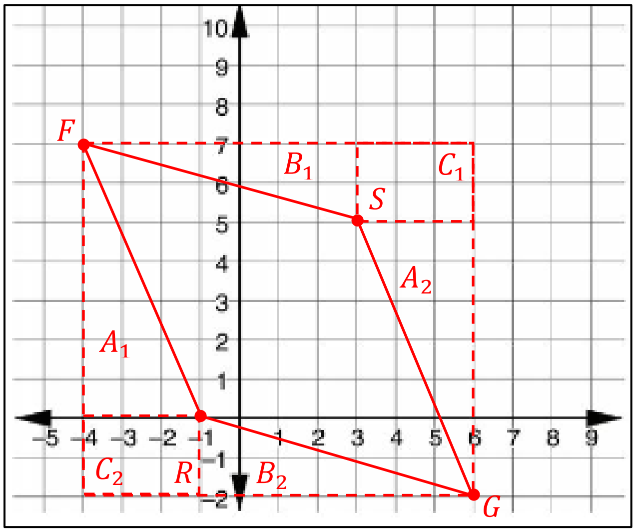
Geometry
Unit 13
Lesson 13 Practice

Name Answer Key_______
Date___________________
Period_________________

Parallelogram $FRGS$ has vertices at $F(-4,7)$, $R(-1,0)$, $G(6,-2)$, and $S(3,5)$.



1. Graph $FRGS$.

2. Complete the table.

Line Segment	Length
$\overline{FR}$	$FR = \sqrt{7^2 + 3^2} = \sqrt{58}$
$\overline{RG}$	$RG = \sqrt{2^2 + 7^2} = \sqrt{53}$
$\overline{GS}$	$GS = \sqrt{3^2 + 7^2} = \sqrt{58}$
$\overline{FS}$	$FS = \sqrt{7^2 + 2^2} = \sqrt{53}$

3. What is the perimeter of $FRGS$?

$$P = \sqrt{58} + \sqrt{53} + \sqrt{58} + \sqrt{53} = 2\sqrt{58} + 2\sqrt{53} \approx 29.79$$

4. What is the area of $FRGS$?

$A_{TOTAL} = (9)(10) = 90$

$\therefore A_{FRGS} = 90 - 2(10.5 + 7 + 6)$
$= 43 \text{ units}^2$

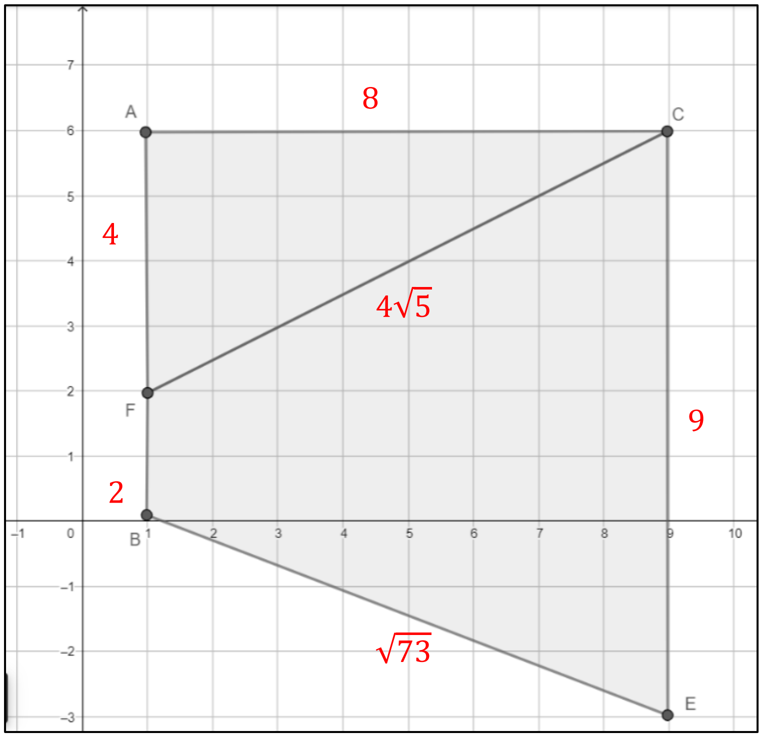
Area of Regions:

$A_1 = A_2 = \frac{1}{2}(7)(3)$
$= 10.5$

$B_1 = B_2 = \frac{1}{2}(7)(2)$
$= 7$

$C_1 = C_2 = (3)(2)$
$= 6$

The composite figure shown is created by $A(1,6)$, $B(1,0)$, $C(9,6)$, $F(1,2)$, and $E(9,-3)$.



5. Complete the table. Round your answer to the nearest thousandth.

Figure	Perimeter	Area
$\triangle ACF$	$FC = \sqrt{4^2 + 8^2} = \sqrt{80} = 4\sqrt{5}$ $P = 4 + 8 + 4\sqrt{5} \approx 20.944$	$A = \frac{1}{2}(4)(8) = 16$ units^2
$BECF$	$BE = \sqrt{3^2 + 8^2} = \sqrt{73}$ $P = 2 + 4\sqrt{5} + 9 + \sqrt{73} \approx 28.488$	$A = \frac{1}{2}(8)(2 + 9)$ $A = 44$ units^2

6. What is the perimeter of the composite figure?

$P = 8 + 4 + 2 + 9 + \sqrt{73} = 23 + \sqrt{73} \approx 31.544$ units

7. What is the area of the composite figure?

Total Area:
$16 + 44 = 60$ units^2